

IPL Analysis 2008 – 2024

1. Project Overview

This project analyses Indian Premier League (IPL) data from 2008 to 2024, covering matches, players, and team performances. The goal is to uncover insights into match outcomes, player contributions, team strategies, and tournament trends to guide cricket analytics, fan engagement, and sports management decisions.

2. Dataset Summary

- **Datasets Used:**
 - Players Data (batting, bowling styles, positions, images)
 - Matches Data (2008–2024, venues, toss decisions, winners, margins)
 - Teams Data (team IDs, logos, names)
 - Deliveries Data (ball-by-ball, too large to process fully here)
- **Key Features:**
 - Player demographics (bat/bowl style, fielding role)
 - Match details (season, venue, toss, winner, margin, player of match)
 - Team information (logos, short names)
 - Ball-by-ball data (runs, wickets, extras)
- **Data Size:**
 - Matches: ~1,000+ records across seasons
 - Players: 500+ players
 - Teams: 15 franchises
 - Deliveries: millions of rows (granular play-by-play)
- **Missing Data:**
 - Some players lack images or field positions
 - Deliveries dataset too large for direct processing

3. Exploratory Data Analysis (Python)

1. Dataset Inspection

- Imported the IPL **matches** and **deliveries** datasets using Pandas.
- Examined the first few records of both datasets using **head()**.
- Checked the **number of rows and columns** using **shape**.
- Used **info()** to understand:
 - Column names
 - Data types
 - Number of non-null values
- This helped in understanding the overall structure of the datasets.

2. Missing Value Analysis

- Checked missing values in the matches dataset using **isnull().sum()**.
- Missing values were identified in columns such as:
 - city
 - method
 - player_of_match
 - winner
 - result_margin
 - target_runs
 - target_overs
- Appropriate values were assigned to handle missing data and make the dataset suitable for further analysis.

3. Duplicate Record Analysis

- Checked for duplicate records using **duplicated().sum()**.
- Duplicate records were removed from both the **matches** and **deliveries** datasets.

- This helped avoid duplicate information affecting the analysis.

4. Team Name Analysis and Standardization

- Examined unique values in the winner column.
- Counted the number of unique winning teams.
- Standardized different names representing the same team.
- Examples include:
 - Royal Challengers Bangalore → Royal Challengers Bengaluru
 - Delhi Daredevils → Delhi Capitals
 - Kings XI Punjab → Punjab Kings
 - Rising Pune Supergiants → Rising Pune Supergiant

5. Season-wise Match Analysis

- Analysed the number of matches played in each IPL season.
- Created a count plot to visualize the **number of matches played per season**.
- This helped identify changes in the number of matches across different seasons.

6. Team Performance Analysis

- Calculated the number of matches won by each team.
- Visualized team-wise victories using a bar chart.
- This provided an overview of the historical performance of IPL teams.

7. Toss Decision Analysis

- Analysed the toss_decision column.
- Visualized whether teams preferred to **bat or field** after winning the toss.
- This helped understand toss-related trends.

8. Top Run Scorers Analysis

- Aggregated batsman_runs for each batter.
- Sorted the players according to their total runs.
- Identified the **Top 10 run scorers**.

9. Top Wicket Takers Analysis

- Identified dismissed players from the **deliveries** dataset.
- Excluded **run-out dismissals** from the wicket calculation.
- Grouped dismissals by bowler.
- Identified the **Top 10 wicket-taking bowlers**.
- Visualized the results using a bar chart.

4. Feature Engineering

1. Date-Based Features

The original date column was converted into a proper datetime format.

Three new features were created:

- **Year** – Extracted the year from the match date.
- **Month** – Extracted the month name from the match date.
- **Day** – Extracted the day name from the match date.

2. Weekend Match Feature

A new feature called **weekend_match** was created.

- It identifies whether a match was played on Saturday or Sunday.
- True indicates a weekend match.
- False indicates a weekday match.

3. Toss-to-Match Winner Feature

A new feature called **toss_match_winner** was created.

- It compares the toss_winner and winner columns.

- If both teams are the same, the value is True.
- Otherwise, the value is False.

4. Match Winning Type Feature

A new feature called **win_type** was created based on the match result.

- If the result is **runs**, the match is categorized as Bat First.
- Otherwise, it is categorized as Chase.

5. Total Boundaries Feature

A new feature called **total_boundaries** was created using the deliveries dataset.

- Deliveries resulting in 4 or 6 runs were filtered.
- The number of such deliveries was calculated for each match.
- The resulting boundary count was mapped to the matches dataset using **match_id** and **id**.

This feature represents the total number of fours and sixes hit in each match.

5. Data Analysis using SQL

We performed structured analysis in MySQL to answer key cricket questions:

1. **Most Successful IPL Teams** - Calculated the total number of matches won by each team.

	winner	total_wins
►	Mumbai Indians	144
	Chennai Super Kings	138
	Kolkata Knight Riders	131
	Royal Challengers Bengaluru	123
	Delhi Capitals	115
	Rajasthan Royals	112
	Punjab Kings	112
	Sunrisers Hyderabad	88
	Deccan Chargers	29
	Gujarat Titans	28
	Lucknow Super Giants	24
	Rising Pune Supergiant	15
	Gujarat Lions	13
	Pune Warriors	12
	Kochi Tuskers Kerala	6

- 2. Player of the Match Analysis** - Counted the number of top 10 Player of the Match awards received by each player.

	player_of_match	award
►	AB de Villiers	25
	CH Gayle	22
	RG Sharma	19
	V Kohli	18
	DA Warner	18
	MS Dhoni	17
	SR Watson	16
	YK Pathan	16
	RA Jadeja	16
	AD Russell	15

- 3. Top 10 Run Scorers** - Calculated the total runs scored by each batter using SUM(batsman_runs).

	batter	total_runs
▶	V Kohli	8014
	S Dhawan	6769
	RG Sharma	6630
	DA Warner	6567
	SK Raina	5536
	MS Dhoni	5243
	AB de Villiers	5181
	CH Gayle	4997
	RV Uthappa	4954
	KD Karthik	4843

- 4. Top 10 Wicket Takers** - Used the player_dismissed column to identify dismissals.

	bowler	wicket
▶	YS Chahal	213
	DJ Bravo	207
	PP Chawla	201
	SP Narine	200
	R Ashwin	198
	B Kumar	195
	SL Malinga	188
	A Mishra	183
	JJ Bumrah	182
	RA Jadeja	169

- 5. Venue-wise Match Analysis** - Counted the number of IPL matches played at each venue.

	venue	match_played
▶	Eden Gardens	77
	Wankhede Stadium	73
	M Chinnaswamy Stadium	65
	Feroz Shah Kotla	60
	Rajiv Gandhi International Stadium, Uppal	49
	MA Chidambaram Stadium, Chepauk	48
	Sawai Mansingh Stadium	47
	Dubai International Cricket Stadium	46
	Wankhede Stadium, Mumbai	45
	Punjab Cricket Association Stadium, Mohali	35
	Sheikh Zayed Stadium	29
	Sharjah Cricket Stadium	28
	MA Chidambaram Stadium, Chepauk, Che...	28
	Narendra Modi Stadium, Ahmedabad	24
	Maharashtra Cricket Association Stadium	22

6. Toss Impact Analysis - Compared the toss_winner with the actual winner.

	toss_impact	total_matches
▶	lost_match	672
	win_match	423

7. Highest Team Scores - Calculated the total score for each team in each innings.

	match_id	inning	batting_team	total_score
▶	1426268	1	Sunrisers Hyderabad	287
	1422126	1	Sunrisers Hyderabad	277
	1422134	1	Kolkata Knight Riders	272
	1426273	1	Sunrisers Hyderabad	266
	598027	1	Royal Challengers Bangalore	263
	1426280	2	Punjab Kings	262
	1426268	2	Royal Challengers Bengaluru	262
	1426280	1	Kolkata Knight Riders	261
	1426281	1	Delhi Capitals	257
	1359512	1	Lucknow Super Giants	257

8. Most Sixes by Batters - Filtered deliveries where batsman_runs = 6.

	batter	total_sixes
▶	CH Gayle	359
	RG Sharma	281
	V Kohli	273
	AB de Villiers	253
	MS Dhoni	252
	DA Warner	236
	KA Pollard	224
	AD Russell	209
	SV Samson	206
	SK Raina	204

9. Season-wise Champions - Selected the season and winning team from the matches table.

	season	winner
▶	2008	Rajasthan Royals
	2009	Deccan Chargers
	2010	Chennai Super Kings
	2011	Chennai Super Kings
	2012	Kolkata Knight Riders
	2013	Mumbai Indians
	2014	Kolkata Knight Riders
	2015	Mumbai Indians
	2016	Sunrisers Hyderabad
	2017	Mumbai Indians
	2018	Chennai Super Kings
	2019	Mumbai Indians
	2020	Mumbai Indians
	2021	Chennai Super Kings
	2022	Gujarat Titans

10. Player Ranking Based on Total Runs - Created a CTE (Common Table Expression) named batter_runs.

	player_rank	batter	total_runs
▶	1	V Kohli	8014
	2	S Dhawan	6769
	3	RG Sharma	6630
	4	DA Warner	6567
	5	SK Raina	5536
	6	MS Dhoni	5243
	7	AB de Villiers	5181
	8	CH Gayle	4997
	9	RV Uthappa	4954
	10	KD Karthik	4843
	11	KL Rahul	4689
	12	AM Rahane	4642
	13	F du Plessis	4571
	14	SV Samson	4419
	15	AT Rayudu	4348

11. High Strike Rate Batters - Calculated total runs and balls played for each batter.

	batter	total_runs	ball_played	strike_rate
▶	AD Russell	2488	1515	164.22
	H Klaasen	993	613	161.99
	SP Narine	1534	984	155.89
	N Pooran	1769	1143	154.77
	LS Livingstone	939	609	154.19
	GJ Maxwell	2772	1842	150.49
	RM Patidar	799	534	149.63
	Abhishek Sharma	1377	925	148.86
	V Sehwag	2728	1833	148.83
	AB de Villiers	5181	3487	148.58

12. Most Economical Bowlers - Calculated total runs conceded by each bowler.

	bowler	runs_given	ball_bowled	bowler_economy
▶	A Kumble	1089	983	6.65
	M Muralitharan	1765	1581	6.70
	SP Narine	4672	4146	6.76
	DW Steyn	2583	2282	6.79
	DL Vettori	894	785	6.83
	Rashid Khan	3340	2901	6.91
	J Botha	818	709	6.92
	R Ashwin	5435	4679	6.97
	MM Ali	900	770	7.01
	SL Malinga	3486	2974	7.03

6. Dashboard in Power BI

Power BI Dashboard Features:

- Created interactive KPI cards for important IPL statistics.
- Added season-wise filtering using slicers.
- Created season-wise Orange Cap and Purple Cap analysis.
- Visualized team points tables.
- Added player-level analysis for fours and sixes.
- Used player images and team logos to improve dashboard usability.
- Created interactive visuals to allow users to explore different IPL seasons.



💡 7. Business & Sports Recommendations

- **Team Strategy:** Focus on toss decisions to win maximum matches.
- **Player Retention:** Retain consistent performers (e.g., M.S Dhoni, Rohit Sharma, Virat Kohli).
- **Fan Engagement:** Highlight rivalries (MI vs CSK) in marketing campaigns.
- **Performance Analytics:** Use ball-by-ball data to identify clutch performers.
- **Tournament Growth:** Promote high-scoring matches and close finishes to boost viewership.

🔧 8. Tools & Technologies Used

Tool / Technology	Purpose
Python	Used for data preprocessing, cleaning, feature engineering, and exploratory data analysis (EDA).
Pandas	Performed data manipulation, handled missing values, transformed columns, and prepared the dataset for analysis.

SQL	Used to Filtering , Grouping , Sorting , Aggregation , Result limitation , Common Table Expression
Power BI	Created an interactive dashboard to visualize Team winners , Match played , Orange cap holder , Purple cap holder
Jupyter Notebook	Developed and executed Python scripts for data cleaning, preprocessing, and analysis.

9. Conclusion

This project successfully analysed IPL data from 2008–2024, revealing trends in team strategies, player performances, and match outcomes. The analysis followed a complete data analytics workflow:

- **Python** for preprocessing,
- **SQL** for structured queries,
- **Power BI** for visualization.

Key insights include the dominance of MI and CSK, the evolving impact of toss decisions, the rise of aggressive batting post-2015, and the importance of star players in shaping match outcomes. These findings can guide franchises, analysts, and fans in understanding the dynamics of IPL and making data-driven decisions.